BIOCHEMISTRY CUMULATIVE EXAM
November 14, 1998

Information transfer molecular biology.

1. The central dogma of molecular biology has said that information flows from DNA to RNA to protein. How did each of the following discoveries make the initially simple picture more complicated?

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   a. (5 points) reverse transcriptase.
   b. (5 points) introns.

1. (10 points) Why is DNA resistant to chemical hydrolysis by alkali, whereas RNA is not? Use a reaction mechanism as part of your explanation.

2. (10 points) Explain one or two biological advantages of having chromosomes composed of double-stranded DNA as opposed to single-stranded DNA.

3. Define the following:

   a. (5 points) point mutation.

   b. (5 points) frameshift mutation.

   c. (5 points) nonsense mutation.

   d. (5 points) insertional mutation with preservation of frame.

5. (10 points) If an enzyme recognizes the sequence AAGGGTT, how many times (on average) would the given sequence be predicted to occur in a genome of 10^{10} nucleotides? (A formula will suffice—final answer from calculator not needed.)

6. (15 points) Site-directed mutagenesis is today an important tool in protein engineering and enzymology. Before 1975, protein chemists wished to be able to perform site-directed mutagenesis, but did not have the proper techniques to do so. What methods were lacking in the "old days.?" Explain three of the important discoveries or developments that have been crucial to the technology that now is commonly used to perform site-directed mutagenesis.

7. (25 points) If you had available a gene for a bovine clotting factor, how would you proceed to find and prepare the equivalent gene from chicken? Explain the experimental steps in as much detail as time and knowledge permit.